Dictionaries and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Dictionaries and records before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Return values and local state

Without opening the old lesson, explain one key idea from Return values and local state. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Lists and indexing

Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Dictionaries and records trace and record the first step you can explain without help.

Use W2: Trace an update for Dictionaries and **record**s. **Record** the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Dictionaries and records explanation against the supplied facts before discussing it.

Use the worked case for Dictionaries and **record**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Read, update and pass a dictionary record

> Predict the three **output**s and trace the named-field update. Explain why zero and a missing count produce different results. If the condition becomes if not count, which supplied **record** is first handled incorrectly?

```
def describe(record):
    count = record.get("count")
    if count is None:
        return "missing"
    return count
record = {"count": 0}
print(describe(record))
record["count"] = 3
print(describe(record))
print(describe({}))
```

My independent explanation, trace and limits

Attempt this case before feedback. **Record** any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Identify fields

> Identify fields. Apply the supplied Dictionaries and records facts and keep your reasoning.

For {"name": "Red", "capacity": 9, "used": 2}, state the keys, value of used and remaining spaces.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace an update

> Trace an update. Apply the supplied Dictionaries and records facts and keep your reasoning.

Change the Red **record** used to 9. State remaining spaces and whether capacity changes.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Repair a lookup

> Repair a lookup. Apply the supplied Dictionaries and records facts and keep your reasoning.

The code reads **record**["Used"] from the Red **record**. Explain the error and repair it without adding a field.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Build a record independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build a record independently. Apply the supplied Dictionaries and records facts and keep your reasoning.

Create a Green **record** with capacity 15 and used 6. Pass its fields to remaining and store the result. Update used to 10 and repeat. Predict both results.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Check the contract

> Check the contract. Apply the supplied Dictionaries and records facts and keep your reasoning.

A **record** has capacity 3 and used 4. It gives -1 spaces without an error. Should it be accepted? Explain.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Dictionaries and records to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled dictionaries and **record**s **evidence** sheet to Python readiness portfolio. Show how you can read and update named **dictionary** fields.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this **record** to last week's **record**; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Dictionaries and records

> A zero field is still present. A dictionary record describes a shelf. The program distinguishes missing data from an actual zero. Explain the labelled step before checking the changed condition.

A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

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Dictionaries and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read a stored field

count exists with numeric value 0.



2. Read an absent field

get("price") returns None because price is missing and no default was supplied.



3. Test key membership

"count" in shelf is True even though its value is zero.

CHANGE ONE THING

Add the field price with value 0.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This **record** is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? "count" in shelf is True even though its value is zero.

1. Test key membership

2. Read an absent field

3. Read a stored field

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or **record**s it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Test key membership "count" in shelf is True even though its value is zero. **Dictionary** keys name fields.

Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add the field price with value 0. Which conclusion is supported by the supplied facts?

1. The price lookup now returns 0 instead of None.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The price lookup now returns 0 instead of None. A **record**ed zero is a present value, whereas the earlier lookup had no field.

Repair a missing building block

Review Return values and local state (python-bridge/9). Goal: Use a returned value in a later calculation. Short practice: Trace the two returned values and explain why first stays seven.

What happens if the caller tries to print result directly?. Return to this week and explain what you changed.

Review Lists and indexing (python-bridge/7). Goal: Distinguish a list index from the value stored at that position. Short practice: Label the three index positions.

Predict all original **output**s, then name the first **output** and stopping point after using index 3.. Return to this week and explain what you changed.

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16.

Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Dictionaries and records picture from its labels, then record a relevant limit.

01. Dictionaries and records: the case

AI ACADEMY / PYTHON-BRIDGE/10 · 01 · LOOK AT THE FACTS

Dictionaries and records: the case

The situation

A zero field is still present. A dictionary record describes a shelf. The program distinguishes missing data from an actual zero.

Supplied evidence

Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

Trace dictionaries and records and explain the changed program before running it.

Follow the labelled picture with the matching explanation. Open the vector diagram

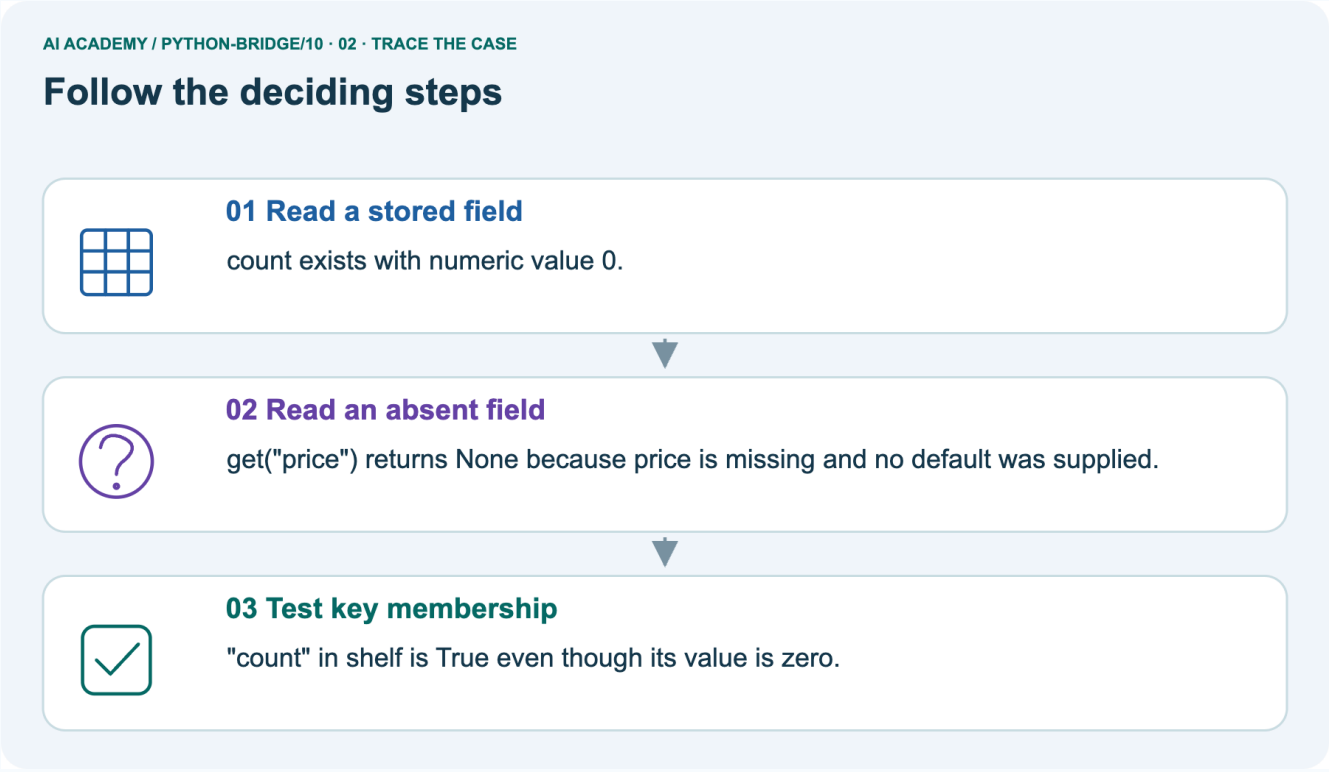
A zero field is still present. A **dictionary record** describes a shelf. The program distinguishes missing data from an actual zero.

Supplied facts: Python 3: shelf={"count":0,"name":"Atlas"}; print(shelf["count"]); print(shelf.get("price")); print("count" in shelf).

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read a stored field: count exists with numeric value 0. → Read an absent field: get("price") returns None because price is missing and no default was supplied. → Test key membership: "count" in shelf is True even though its value is zero.

Point to each numbered stage and explain which **record**ed **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence

AI ACADEMY / PYTHON-BRIDGE/10 · 03 · INSPECT THE LESSON MODEL

Follow the evidence

Step or case	Evidence	Meaning
Read used	4	12 - 4 = 8
Update used	6	12 - 6 = 6

Read each row from left to right and connect the evidence to the stated rule.

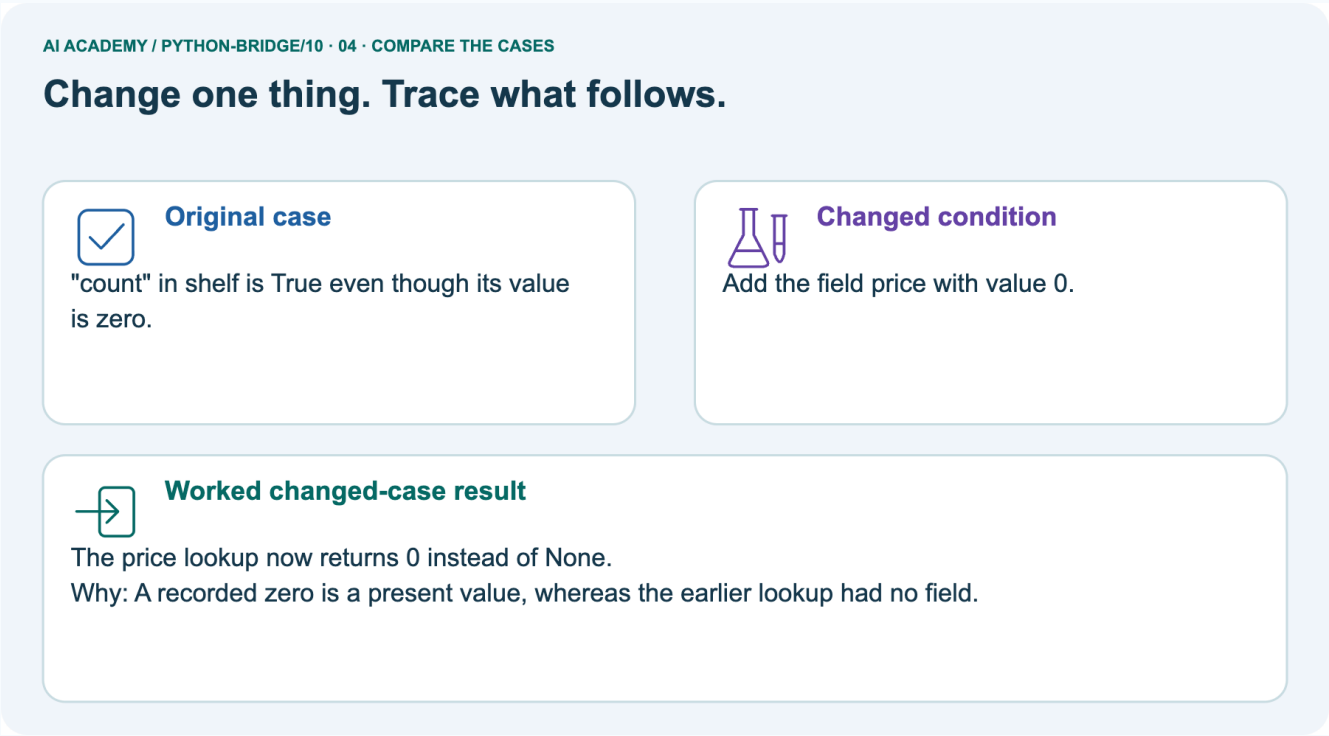
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Read used | 4 | 12 - 4 = 8. Update used | 6 | 12 - 6 = 6

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

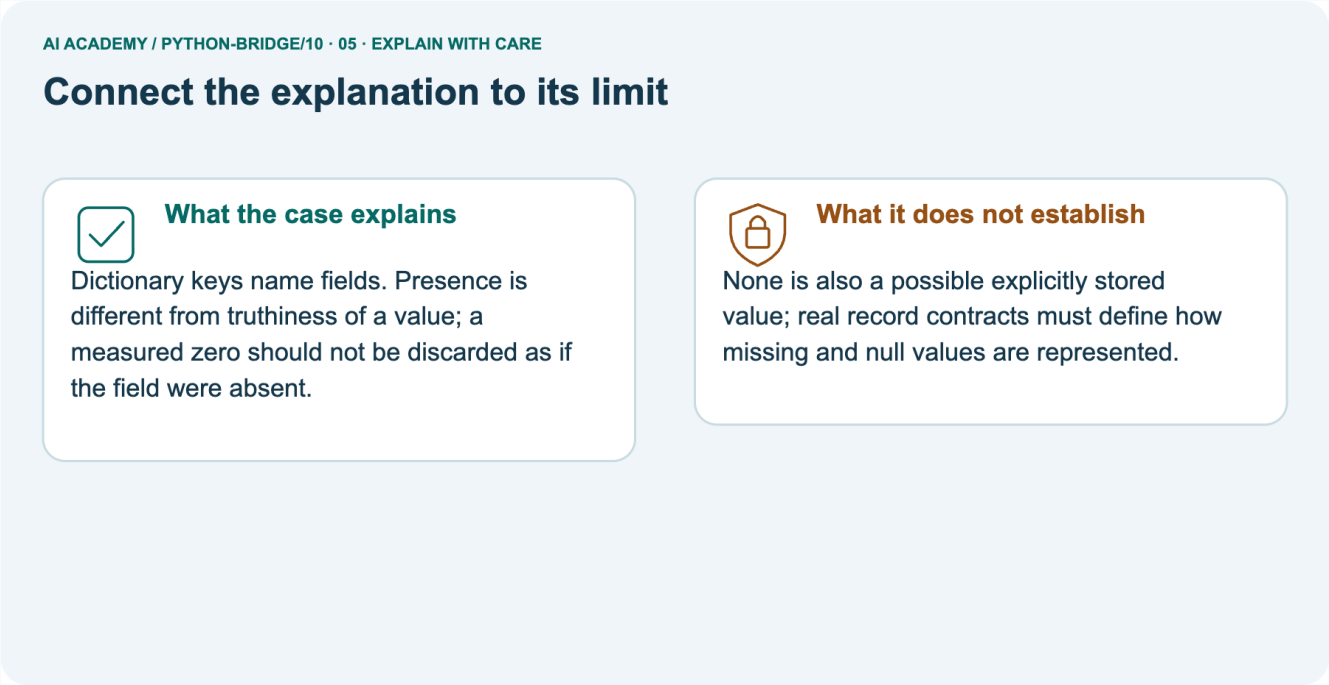
Original case: "count" in shelf is True even though its value is zero. Changed condition: Add the field price with value 0. Expected result: The price lookup now returns 0 instead of None.

Reason: A **record**ed zero is a present value, whereas the earlier lookup had no field.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

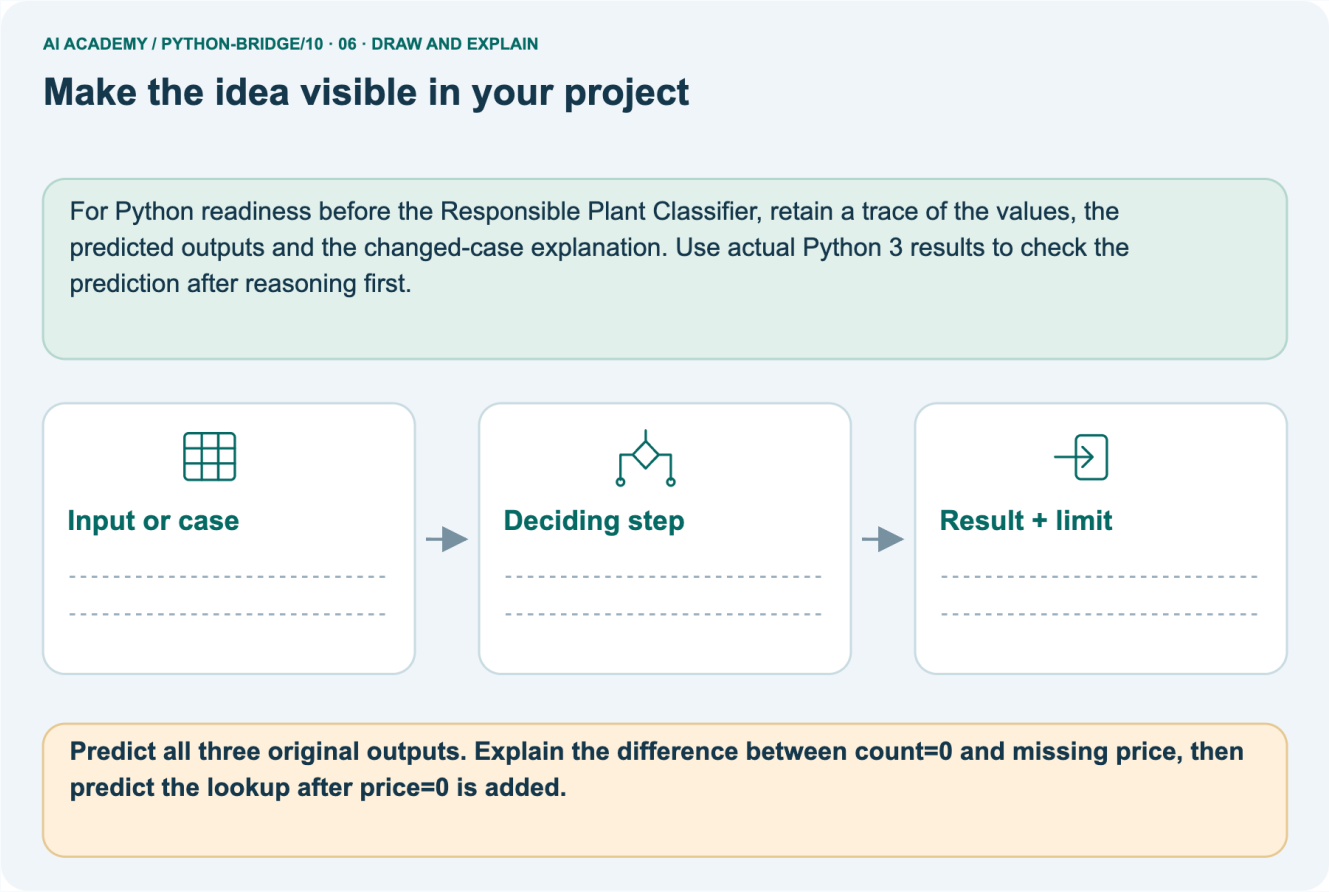
Dictionary keys name fields. Presence is different from truthiness of a value; a measured zero should not be discarded as if the field were absent.

Limit: None is also a possible explicitly stored value; real **record** contracts must define how missing and null values are represented.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Predict all three original **output**s. Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.